

Supporting Information for: Validation Noise, Not the Optimizer, Limits Metaheuristic Tuning in Molecular Property Prediction

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List of Figures

Figure S1. Sphere-function verification of the optimizer implementations.	2
Figure S2. Pairwise Vargha–Delaney A12 between optimizers on the primary test metric.....	3
Figure S3. Critical-difference diagrams across datasets for the five optimizers (low-powered).....	4
Figure S4. Sensitivity of GWO and WOA to the coefficient schedule: a_0 versus reference-faithful variants.....	5
Figure S5. Convergence of the best-so-far validation fitness.....	6

List of Tables

Table S1. Tuning versus the default configuration, all test metrics.	7
Table S2. Friedman tests across the five optimizers (default excluded).	10
Table S3. Pairwise comparisons of optimizers on the primary metric: Wilcoxon signed-rank test with Holm correction and Vargha–Delaney A12.	11
Table S4. Validation–test gap and Spearman correlation between best validation and test scores.....	13
Table S5. Selected-feature fraction and Nogueira stability (Experiment B).	14
Table S6. Mirrored-encoding robustness check (pre-registered tests, Experiment B).	15
Table S7. Sensitivity analysis: a_0 versus reference-faithful coefficient schedule for GWO and WOA.....	16
Table S8. Compute cost per dataset and optimizer.	17
Table S9. Validation headroom versus test gain (exploratory, not pre-specified).	18
Table S10. Data-curation check: duplicate molecules within and across the scaffold splits.....	19

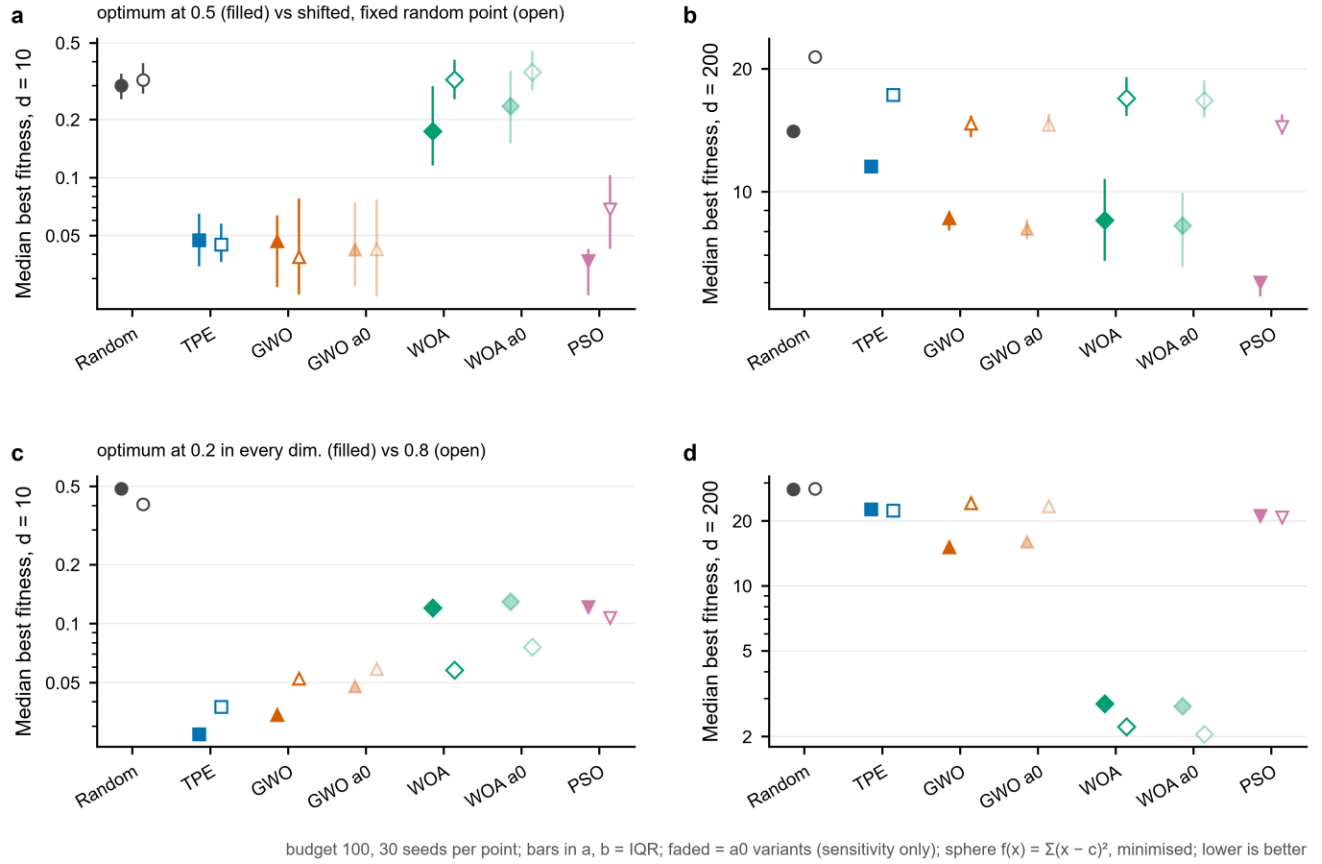


Figure S1. Sphere-function verification of the optimizer implementations.

Median best fitness after 100 evaluations over 30 seeds (lower is better). (a, b) Optimum at 0.5 in every dimension (filled markers) versus a shifted optimum at a fixed random point in $[0.1, 0.9]^d$ (open markers), for $d = 10$ and $d = 200$; bars show the interquartile range. (c, d) Optimum at 0.2 (filled) versus 0.8 (open) in every dimension. Faded markers: the a0 variants of GWO and WOA (sensitivity analysis only). At $d = 200$ with the shifted optimum, PSO (14.4) and GWO (14.7) reach the lowest values and Random the highest (21.4). GWO does worse when the optimum is at 0.8 than at 0.2 (ratio 1.60 at $d = 200$; 1.46 for the a0 variant), consistent with a bias toward $x = 0$, whereas Random, TPE and PSO show ratios of 0.986–1.01. WOA reaches unusually low values when the optimum lies on the diagonal (2.84 and 2.22 at $d = 200$).

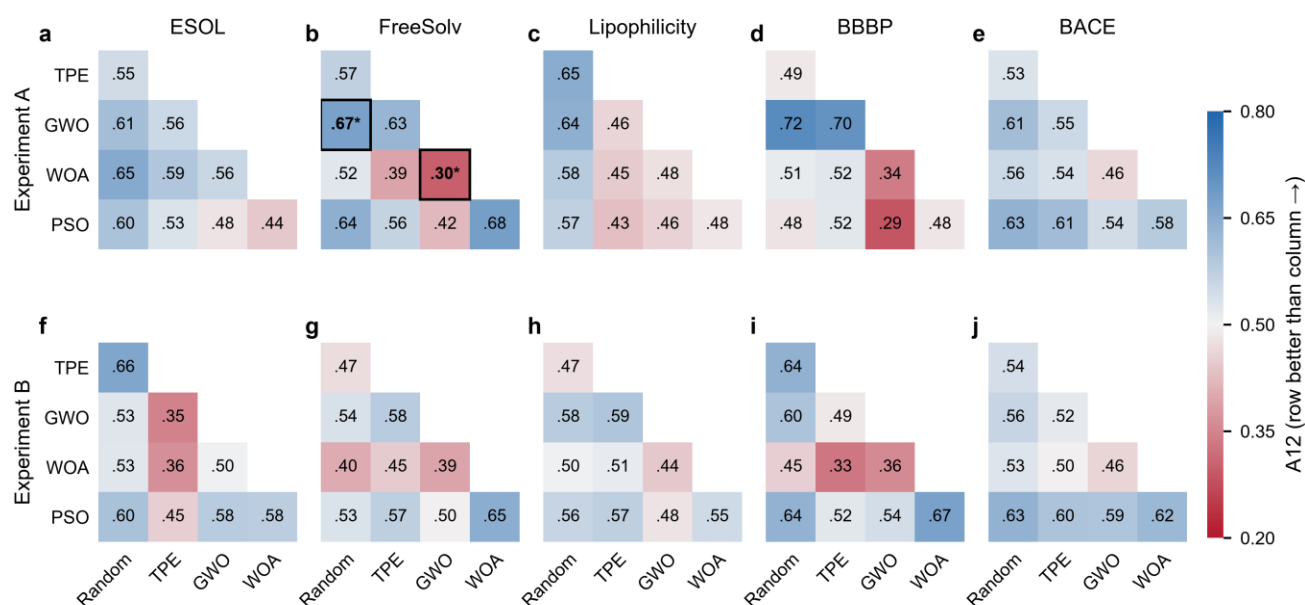
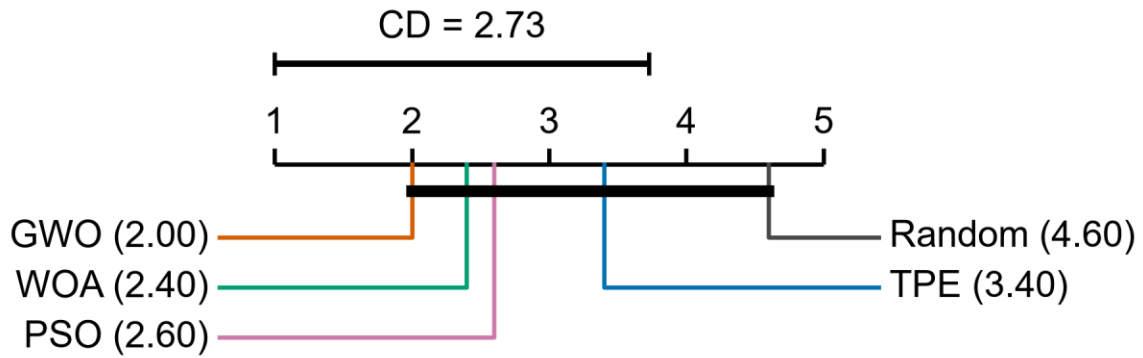


Figure S2. Pairwise Vargha–Delaney A12 between optimizers on the primary test metric.

Lower triangle: A12 of the row optimizer versus the column optimizer, that is, the probability that the row optimizer scores better on the test set (0.5 = no difference), from 30 seeds per method, for each dataset (columns) and experiment (rows). Boxed cells with an asterisk are significant after Holm correction over the 10 pairs (paired Wilcoxon signed-rank test, zero_method = pratt). Only 2 of the 100 comparisons are significant, both for FreeSolv in Experiment A: GWO outperforms Random (cell GWO vs Random, A12 = 0.67) and WOA (cell WOA vs GWO, A12 = 0.30, i.e., 0.70 in favour of GWO). All values are listed in Table S3.

- a** Experiment A: optimizers, mean rank over N = 5 datasets
LOW-POWERED; Nemenyi CD at $\alpha = 0.05$; Friedman $p = 0.08$



- b** Experiment B: optimizers, mean rank over N = 5 datasets
LOW-POWERED; Nemenyi CD at $\alpha = 0.05$; Friedman $p = 0.02$

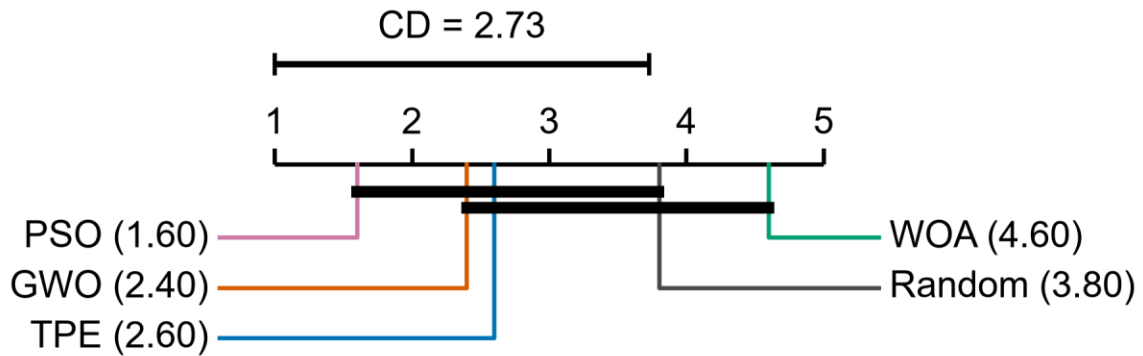


Figure S3. Critical-difference diagrams across datasets for the five optimizers (low-powered).

Mean rank of each optimizer across the N = 5 datasets (rank 1 = best median test score on the primary metric), with the Nemenyi critical difference at $\alpha = 0.05$ (CD = 2.73); thick bars join optimizers whose mean ranks differ by less than the CD. (a) Experiment A: GWO 2.00, WOA 2.40, PSO 2.60, TPE 3.40, Random 4.60; Friedman $p = 0.075$; no pair differs by more than the CD. (b) Experiment B: PSO 1.60, GWO 2.40, TPE 2.60, Random 3.80, WOA 4.60; Friedman $p = 0.023$; only PSO and WOA differ by more than the CD (3.00). With five datasets these tests have low power; the per-dataset analyses (Tables S2 and S3) are the primary evidence.

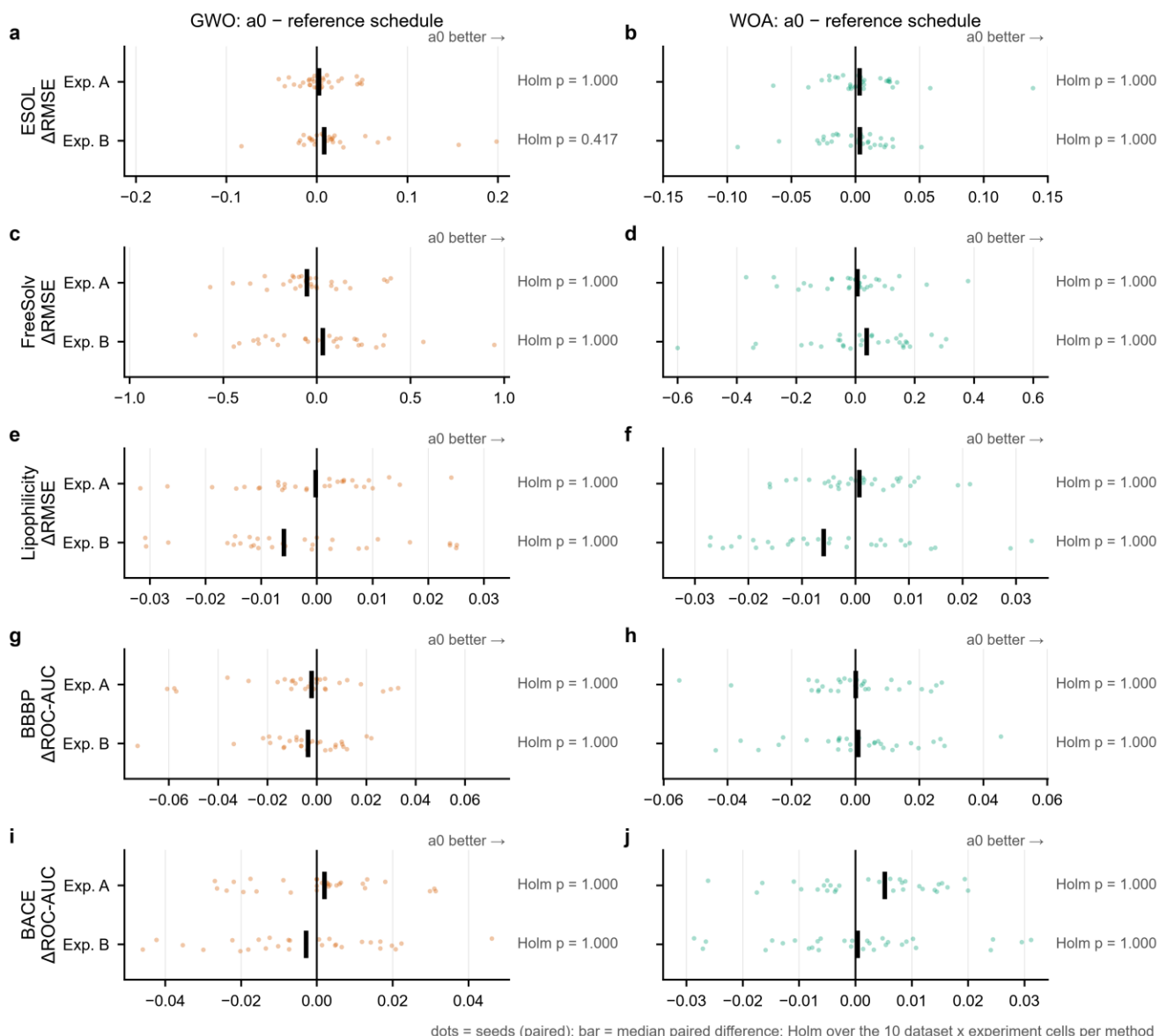


Figure S4. Sensitivity of GWO and WOA to the coefficient schedule: a0 versus reference-faithful variants.

Paired differences ($a_0 - \text{reference}$) in the primary test metric for each of the 30 seeds, oriented so that positive values favour the a_0 schedule; bars mark the median paired difference. Rows are datasets, columns are GWO and WOA, and each panel shows Experiments A and B. Holm-adjusted p-values (paired Wilcoxon test, Holm correction over the 10 dataset $\times$ experiment cells per method) are shown at right; none is significant (smallest Holm $p = 0.417$, GWO, ESOL, Experiment B). The a_0 schedule reduces the median number of unique evaluations per run (GWO 91 versus 100; WOA 91.5–93 versus 96–97.5) but changes none of the 50 tuning-versus-default conclusions. Values in Table S7.

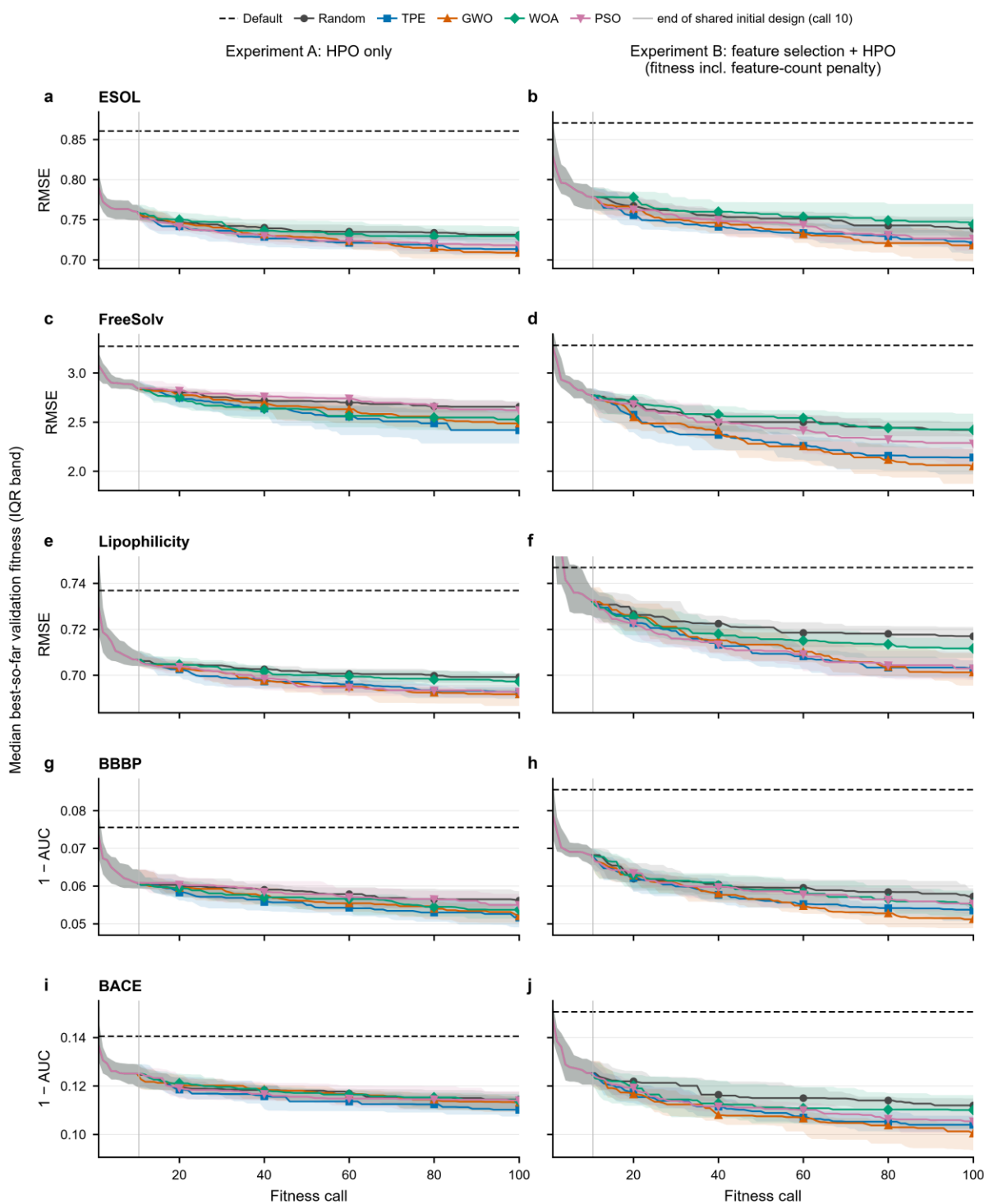


Figure S5. Convergence of the best-so-far validation fitness.

Median (line) and interquartile range (band) over 30 seeds of the best-so-far validation fitness against the fitness-call index (1–100), per dataset (rows) and experiment (columns): RMSE for regression and $1 - \text{ROC-AUC}$ for classification; in Experiment B the fitness includes the feature-count penalty ($0.01 \times \text{selected fraction}$). Dashed line: validation fitness of the default configuration. The grey vertical line marks the end of the shared 10-point initial design (calls 1–10 are identical for all optimizers). For every optimizer, dataset and experiment, the median best-so-far fitness is already below the default's by call 10.

Table S1. Tuning versus the default configuration, all test metrics.

Dataset	Exp.	Metric	Method	Default	Median [95% CI]	Improvement [95% CI]	% seeds better	p (Holm)	Conclusion
ESOL	A	RMSE (primary)	Random	0.889	0.924 [0.914, 0.931]	-0.0351 [-0.0418, -0.0253]	3.33	<0.001	worse
ESOL	A	RMSE (primary)	TPE	0.889	0.919 [0.905, 0.928]	-0.0299 [-0.0394, -0.0163]	6.67	<0.001	worse
ESOL	A	RMSE (primary)	GWO	0.889	0.916 [0.905, 0.929]	-0.0274 [-0.0397, -0.0159]	10.0	<0.001	worse
ESOL	A	RMSE (primary)	WOA	0.889	0.911 [0.896, 0.923]	-0.0221 [-0.0345, -0.00668]	16.7	<0.001	worse
ESOL	A	RMSE (primary)	PSO	0.889	0.916 [0.905, 0.925]	-0.0271 [-0.0363, -0.0158]	6.67	<0.001	worse
ESOL	A	MAE	Random	0.658	0.693 [0.685, 0.703]	-0.0348 [-0.0449, -0.0271]	0	<0.001	worse
ESOL	A	MAE	TPE	0.658	0.690 [0.683, 0.699]	-0.0323 [-0.0411, -0.0252]	0	<0.001	worse
ESOL	A	MAE	GWO	0.658	0.688 [0.680, 0.697]	-0.0297 [-0.0392, -0.0222]	0	<0.001	worse
ESOL	A	MAE	WOA	0.658	0.689 [0.683, 0.693]	-0.0309 [-0.0352, -0.0248]	0	<0.001	worse
ESOL	A	MAE	PSO	0.658	0.691 [0.683, 0.697]	-0.0336 [-0.0387, -0.0253]	3.33	<0.001	worse
ESOL	A	R2	Random	0.793	0.777 [0.773, 0.781]	-0.0166 [-0.0199, -0.0119]	3.33	<0.001	worse
ESOL	A	R2	TPE	0.793	0.779 [0.774, 0.786]	-0.0142 [-0.0188, -0.00768]	6.67	<0.001	worse
ESOL	A	R2	GWO	0.793	0.780 [0.774, 0.786]	-0.0130 [-0.0189, -0.00749]	10.0	<0.001	worse
ESOL	A	R2	WOA	0.793	0.783 [0.777, 0.790]	-0.0104 [-0.0164, -0.00312]	16.7	<0.001	worse
ESOL	A	R2	PSO	0.793	0.780 [0.776, 0.786]	-0.0128 [-0.0172, -0.00742]	6.67	<0.001	worse
ESOL	B	RMSE (primary)	Random	0.889	0.921 [0.898, 0.941]	-0.0321 [-0.0520, -0.00964]	23.3	0.00286	worse
ESOL	B	RMSE (primary)	TPE	0.889	0.900 [0.893, 0.910]	-0.0116 [-0.0215, -0.00383]	30.0	0.0293	worse
ESOL	B	RMSE (primary)	GWO	0.889	0.919 [0.904, 0.935]	-0.0299 [-0.0461, -0.0152]	26.7	<0.001	worse
ESOL	B	RMSE (primary)	WOA	0.889	0.918 [0.897, 0.935]	-0.0291 [-0.0458, -0.00861]	26.7	0.00267	worse
ESOL	B	RMSE (primary)	PSO	0.889	0.905 [0.892, 0.913]	-0.0162 [-0.0241, -0.00348]	30.0	0.00553	worse
ESOL	B	MAE	Random	0.658	0.696 [0.681, 0.710]	-0.0386 [-0.0520, -0.0226]	10.0	<0.001	worse
ESOL	B	MAE	TPE	0.658	0.687 [0.669, 0.693]	-0.0288 [-0.0354, -0.0107]	23.3	<0.001	worse
ESOL	B	MAE	GWO	0.658	0.696 [0.676, 0.706]	-0.0383 [-0.0480, -0.0179]	10.0	<0.001	worse
ESOL	B	MAE	WOA	0.658	0.692 [0.682, 0.720]	-0.0340 [-0.0624, -0.0236]	10.0	<0.001	worse
ESOL	B	MAE	PSO	0.658	0.683 [0.678, 0.693]	-0.0248 [-0.0355, -0.0198]	6.67	<0.001	worse
ESOL	B	R2	Random	0.793	0.778 [0.768, 0.789]	-0.0152 [-0.0249, -0.00451]	23.3	0.00267	worse
ESOL	B	R2	TPE	0.793	0.788 [0.783, 0.791]	-0.00545 [-0.0101, -0.00179]	30.0	0.0234	worse
ESOL	B	R2	GWO	0.793	0.779 [0.771, 0.786]	-0.0142 [-0.0220, -0.00713]	26.7	<0.001	worse
ESOL	B	R2	WOA	0.793	0.779 [0.771, 0.789]	-0.0137 [-0.0219, -0.00403]	26.7	0.00267	worse
ESOL	B	R2	PSO	0.793	0.786 [0.782, 0.792]	-0.00762 [-0.0114, -0.00162]	30.0	0.00512	worse
FreeSolv	A	RMSE (primary)	Random	2.81	2.61 [2.46, 2.73]	0.200 [0.0816, 0.351]	83.3	<0.001	better
FreeSolv	A	RMSE (primary)	TPE	2.81	2.51 [2.48, 2.56]	0.296 [0.254, 0.331]	90.0	<0.001	better
FreeSolv	A	RMSE (primary)	GWO	2.81	2.46 [2.37, 2.52]	0.355 [0.295, 0.443]	96.7	<0.001	better
FreeSolv	A	RMSE (primary)	WOA	2.81	2.60 [2.53, 2.64]	0.208 [0.168, 0.278]	93.3	<0.001	better
FreeSolv	A	RMSE (primary)	PSO	2.81	2.49 [2.42, 2.56]	0.321 [0.246, 0.386]	100	<0.001	better
FreeSolv	A	MAE	Random	1.86	1.73 [1.67, 1.84]	0.136 [0.0247, 0.196]	80.0	<0.001	better
FreeSolv	A	MAE	TPE	1.86	1.70 [1.65, 1.75]	0.165 [0.111, 0.216]	86.7	<0.001	better
FreeSolv	A	MAE	GWO	1.86	1.66 [1.61, 1.68]	0.203 [0.178, 0.258]	93.3	<0.001	better
FreeSolv	A	MAE	WOA	1.86	1.73 [1.67, 1.77]	0.129 [0.0938, 0.194]	90.0	<0.001	better
FreeSolv	A	MAE	PSO	1.86	1.68 [1.64, 1.72]	0.182 [0.142, 0.221]	100	<0.001	better
FreeSolv	A	R2	Random	0.645	0.693 [0.665, 0.728]	0.0487 [0.0203, 0.0832]	83.3	<0.001	better
FreeSolv	A	R2	TPE	0.645	0.716 [0.706, 0.723]	0.0709 [0.0614, 0.0787]	90.0	<0.001	better
FreeSolv	A	R2	GWO	0.645	0.729 [0.715, 0.748]	0.0841 [0.0706, 0.103]	96.7	<0.001	better
FreeSolv	A	R2	WOA	0.645	0.695 [0.686, 0.711]	0.0507 [0.0413, 0.0668]	93.3	<0.001	better
FreeSolv	A	R2	PSO	0.645	0.721 [0.704, 0.736]	0.0765 [0.0595, 0.0910]	100	<0.001	better
FreeSolv	B	RMSE (primary)	Random	2.81	2.46 [2.37, 2.57]	0.355 [0.242, 0.445]	93.3	<0.001	better
FreeSolv	B	RMSE (primary)	TPE	2.81	2.46 [2.38, 2.58]	0.347 [0.228, 0.433]	93.3	<0.001	better
FreeSolv	B	RMSE (primary)	GWO	2.81	2.43 [2.28, 2.52]	0.385 [0.288, 0.527]	90.0	<0.001	better
FreeSolv	B	RMSE (primary)	WOA	2.81	2.54 [2.41, 2.61]	0.267 [0.200, 0.406]	93.3	<0.001	better
FreeSolv	B	RMSE (primary)	PSO	2.81	2.39 [2.34, 2.50]	0.421 [0.306, 0.471]	96.7	<0.001	better
FreeSolv	B	MAE	Random	1.86	1.68 [1.64, 1.78]	0.178 [0.0864, 0.218]	90.0	<0.001	better
FreeSolv	B	MAE	TPE	1.86	1.73 [1.64, 1.76]	0.132 [0.106, 0.225]	93.3	<0.001	better
FreeSolv	B	MAE	GWO	1.86	1.67 [1.63, 1.74]	0.192 [0.118, 0.228]	86.7	<0.001	better
FreeSolv	B	MAE	WOA	1.86	1.76 [1.68, 1.80]	0.0995 [0.0602, 0.187]	83.3	<0.001	better

Dataset	Exp.	Metric	Method	Default	Median [95% CI]	Improvement [95% CI]	% seeds better	p (Holm)	Conclusion
FreeSolv	B	MAE	PSO	1.86	1.71 [1.63, 1.75]	0.154 [0.112, 0.234]	90.0	<0.001	better
FreeSolv	B	R2	Random	0.645	0.729 [0.703, 0.748]	0.0840 [0.0586, 0.104]	93.3	<0.001	better
FreeSolv	B	R2	TPE	0.645	0.727 [0.700, 0.746]	0.0822 [0.0553, 0.101]	93.3	<0.001	better
FreeSolv	B	R2	GWO	0.645	0.735 [0.714, 0.765]	0.0907 [0.0690, 0.121]	90.0	<0.001	better
FreeSolv	B	R2	WOA	0.645	0.709 [0.693, 0.740]	0.0642 [0.0487, 0.0952]	93.3	<0.001	better
FreeSolv	B	R2	PSO	0.645	0.743 [0.718, 0.754]	0.0984 [0.0732, 0.109]	96.7	<0.001	better
Lipophilicity	A	RMSE (primary)	Random	0.760	0.729 [0.726, 0.734]	0.0309 [0.0256, 0.0335]	93.3	<0.001	better
Lipophilicity	A	RMSE (primary)	TPE	0.760	0.725 [0.720, 0.730]	0.0348 [0.0302, 0.0397]	100	<0.001	better
Lipophilicity	A	RMSE (primary)	GWO	0.760	0.726 [0.724, 0.728]	0.0339 [0.0322, 0.0365]	100	<0.001	better
Lipophilicity	A	RMSE (primary)	WOA	0.760	0.727 [0.723, 0.731]	0.0327 [0.0286, 0.0368]	90.0	<0.001	better
Lipophilicity	A	RMSE (primary)	PSO	0.760	0.728 [0.722, 0.730]	0.0316 [0.0300, 0.0377]	90.0	<0.001	better
Lipophilicity	A	MAE	Random	0.581	0.553 [0.549, 0.558]	0.0281 [0.0234, 0.0322]	96.7	<0.001	better
Lipophilicity	A	MAE	TPE	0.581	0.546 [0.540, 0.550]	0.0352 [0.0308, 0.0412]	100	<0.001	better
Lipophilicity	A	MAE	GWO	0.581	0.548 [0.546, 0.549]	0.0334 [0.0317, 0.0351]	100	<0.001	better
Lipophilicity	A	MAE	WOA	0.581	0.549 [0.545, 0.552]	0.0323 [0.0294, 0.0364]	93.3	<0.001	better
Lipophilicity	A	MAE	PSO	0.581	0.549 [0.546, 0.554]	0.0318 [0.0275, 0.0349]	100	<0.001	better
Lipophilicity	A	R2	Random	0.601	0.632 [0.627, 0.635]	0.0318 [0.0264, 0.0345]	93.3	<0.001	better
Lipophilicity	A	R2	TPE	0.601	0.636 [0.632, 0.641]	0.0357 [0.0312, 0.0406]	100	<0.001	better
Lipophilicity	A	R2	GWO	0.601	0.635 [0.634, 0.638]	0.0348 [0.0331, 0.0374]	100	<0.001	better
Lipophilicity	A	R2	WOA	0.601	0.634 [0.630, 0.638]	0.0336 [0.0295, 0.0377]	90.0	<0.001	better
Lipophilicity	A	R2	PSO	0.601	0.633 [0.632, 0.639]	0.0325 [0.0309, 0.0387]	90.0	<0.001	better
Lipophilicity	B	RMSE (primary)	Random	0.760	0.738 [0.735, 0.742]	0.0222 [0.0182, 0.0250]	93.3	<0.001	better
Lipophilicity	B	RMSE (primary)	TPE	0.760	0.738 [0.736, 0.745]	0.0218 [0.0154, 0.0245]	93.3	<0.001	better
Lipophilicity	B	RMSE (primary)	GWO	0.760	0.734 [0.729, 0.742]	0.0260 [0.0183, 0.0312]	93.3	<0.001	better
Lipophilicity	B	RMSE (primary)	WOA	0.760	0.739 [0.730, 0.749]	0.0209 [0.0111, 0.0303]	93.3	<0.001	better
Lipophilicity	B	RMSE (primary)	PSO	0.760	0.734 [0.731, 0.742]	0.0257 [0.0180, 0.0293]	93.3	<0.001	better
Lipophilicity	B	MAE	Random	0.581	0.559 [0.555, 0.562]	0.0222 [0.0189, 0.0261]	93.3	<0.001	better
Lipophilicity	B	MAE	TPE	0.581	0.555 [0.551, 0.563]	0.0256 [0.0181, 0.0303]	96.7	<0.001	better
Lipophilicity	B	MAE	GWO	0.581	0.557 [0.551, 0.560]	0.0238 [0.0211, 0.0306]	100	<0.001	better
Lipophilicity	B	MAE	WOA	0.581	0.555 [0.549, 0.568]	0.0259 [0.0127, 0.0322]	90.0	<0.001	better
Lipophilicity	B	MAE	PSO	0.581	0.553 [0.550, 0.561]	0.0278 [0.0203, 0.0310]	96.7	<0.001	better
Lipophilicity	B	R2	Random	0.601	0.624 [0.620, 0.626]	0.0230 [0.0189, 0.0258]	93.3	<0.001	better
Lipophilicity	B	R2	TPE	0.601	0.623 [0.617, 0.626]	0.0225 [0.0161, 0.0253]	93.3	<0.001	better
Lipophilicity	B	R2	GWO	0.601	0.628 [0.620, 0.633]	0.0269 [0.0190, 0.0321]	93.3	<0.001	better
Lipophilicity	B	R2	WOA	0.601	0.622 [0.612, 0.632]	0.0216 [0.0116, 0.0312]	93.3	<0.001	better
Lipophilicity	B	R2	PSO	0.601	0.627 [0.619, 0.631]	0.0266 [0.0187, 0.0302]	93.3	<0.001	better
BBBP	A	ROC-AUC (primary)	Random	0.915	0.903 [0.896, 0.906]	-0.0121 [-0.0193, -0.00887]	3.33	<0.001	worse
BBBP	A	ROC-AUC (primary)	TPE	0.915	0.900 [0.895, 0.906]	-0.0148 [-0.0204, -0.00900]	16.7	<0.001	worse
BBBP	A	ROC-AUC (primary)	GWO	0.915	0.909 [0.907, 0.914]	-0.00569 [-0.00853, -0.000875]	26.7	0.00115	worse
BBBP	A	ROC-AUC (primary)	WOA	0.915	0.904 [0.895, 0.908]	-0.0117 [-0.0202, -0.00706]	16.7	<0.001	worse
BBBP	A	ROC-AUC (primary)	PSO	0.915	0.901 [0.898, 0.906]	-0.0142 [-0.0167, -0.00906]	10.0	<0.001	worse
BBBP	A	PR-AUC	Random	0.962	0.958 [0.956, 0.961]	-0.00366 [-0.00630, -0.00118]	23.3	0.00566	worse
BBBP	A	PR-AUC	TPE	0.962	0.960 [0.952, 0.962]	-0.00241 [-0.0105, -0.000167]	33.3	0.0116	worse
BBBP	A	PR-AUC	GWO	0.962	0.964 [0.960, 0.965]	0.00162 [-0.00243, 0.00306]	60.0	0.919	n.s.
BBBP	A	PR-AUC	WOA	0.962	0.959 [0.951, 0.962]	-0.00339 [-0.0108, -0.000299]	30.0	0.00566	worse
BBBP	A	PR-AUC	PSO	0.962	0.959 [0.952, 0.962]	-0.00262 [-0.00952, -0.000353]	26.7	0.00896	worse
BBBP	B	ROC-AUC (primary)	Random	0.915	0.904 [0.896, 0.911]	-0.0112 [-0.0192, -0.00462]	13.3	<0.001	worse
BBBP	B	ROC-AUC (primary)	TPE	0.915	0.911 [0.904, 0.914]	-0.00394 [-0.0116, -0.00150]	16.7	0.00249	worse
BBBP	B	ROC-AUC (primary)	GWO	0.915	0.907 [0.902, 0.914]	-0.00818 [-0.0128, -0.001000]	26.7	0.00267	worse
BBBP	B	ROC-AUC (primary)	WOA	0.915	0.904 [0.894, 0.908]	-0.0115 [-0.0211, -0.00668]	10.0	<0.001	worse
BBBP	B	ROC-AUC (primary)	PSO	0.915	0.909 [0.903, 0.915]	-0.00594 [-0.0121, -0.000375]	30.0	0.00316	worse
BBBP	B	PR-AUC	Random	0.962	0.962 [0.956, 0.963]	-0.000438 [-0.00578, 0.00142]	46.7	0.524	n.s.
BBBP	B	PR-AUC	TPE	0.962	0.963 [0.959, 0.964]	0.000700 [-0.00282, 0.00212]	60.0	1.00	n.s.
BBBP	B	PR-AUC	GWO	0.962	0.964 [0.960, 0.966]	0.00170 [-0.00222, 0.00365]	60.0	1.00	n.s.
BBBP	B	PR-AUC	WOA	0.962	0.961 [0.957, 0.964]	-0.000831 [-0.00520, 0.00179]	46.7	0.657	n.s.
BBBP	B	PR-AUC	PSO	0.962	0.964 [0.960, 0.966]	0.00152 [-0.00206, 0.00374]	56.7	1.00	n.s.

Dataset	Exp.	Metric	Method	Default	Median [95% CI]	Improvement [95% CI]	% seeds better	p (Holm)	Conclusion
BACE	A	ROC-AUC (primary)	Random	0.814	0.810 [0.805, 0.815]	-0.00433 [-0.00990, 0.0000962]	33.3	0.351	n.s.
BACE	A	ROC-AUC (primary)	TPE	0.814	0.814 [0.806, 0.818]	-0.000769 [-0.00808, 0.00356]	46.7	1.00	n.s.
BACE	A	ROC-AUC (primary)	GWO	0.814	0.815 [0.810, 0.818]	0.000192 [-0.00452, 0.00385]	53.3	1.00	n.s.
BACE	A	ROC-AUC (primary)	WOA	0.814	0.816 [0.809, 0.821]	0.00135 [-0.00529, 0.00635]	53.3	1.00	n.s.
BACE	A	ROC-AUC (primary)	PSO	0.814	0.819 [0.811, 0.824]	0.00471 [-0.00308, 0.00913]	63.3	1.00	n.s.
BACE	A	PR-AUC	Random	0.709	0.703 [0.696, 0.705]	-0.00671 [-0.0138, -0.00414]	23.3	0.0215	worse
BACE	A	PR-AUC	TPE	0.709	0.694 [0.686, 0.705]	-0.0151 [-0.0234, -0.00399]	30.0	0.00792	worse
BACE	A	PR-AUC	GWO	0.709	0.702 [0.694, 0.709]	-0.00726 [-0.0151, -0.000478]	33.3	0.0493	worse
BACE	A	PR-AUC	WOA	0.709	0.705 [0.701, 0.712]	-0.00398 [-0.00799, 0.00252]	33.3	0.382	n.s.
BACE	A	PR-AUC	PSO	0.709	0.707 [0.699, 0.713]	-0.00241 [-0.0101, 0.00333]	46.7	0.382	n.s.
BACE	B	ROC-AUC (primary)	Random	0.814	0.813 [0.808, 0.823]	-0.00106 [-0.00644, 0.00808]	46.7	1.00	n.s.
BACE	B	ROC-AUC (primary)	TPE	0.814	0.818 [0.808, 0.823]	0.00356 [-0.00635, 0.00865]	53.3	1.00	n.s.
BACE	B	ROC-AUC (primary)	GWO	0.814	0.819 [0.813, 0.823]	0.00413 [-0.00183, 0.00808]	56.7	0.869	n.s.
BACE	B	ROC-AUC (primary)	WOA	0.814	0.812 [0.809, 0.823]	-0.00288 [-0.00548, 0.00894]	46.7	1.00	n.s.
BACE	B	ROC-AUC (primary)	PSO	0.814	0.820 [0.815, 0.829]	0.00529 [0.0000962, 0.0141]	66.7	0.106	n.s.
BACE	B	PR-AUC	Random	0.709	0.696 [0.692, 0.720]	-0.0132 [-0.0177, 0.0101]	40.0	0.914	n.s.
BACE	B	PR-AUC	TPE	0.709	0.709 [0.698, 0.716]	-0.000959 [-0.0115, 0.00622]	46.7	0.914	n.s.
BACE	B	PR-AUC	GWO	0.709	0.705 [0.693, 0.714]	-0.00458 [-0.0169, 0.00405]	43.3	0.914	n.s.
BACE	B	PR-AUC	WOA	0.709	0.705 [0.699, 0.712]	-0.00442 [-0.0106, 0.00264]	36.7	0.790	n.s.
BACE	B	PR-AUC	PSO	0.709	0.704 [0.697, 0.716]	-0.00504 [-0.0120, 0.00661]	43.3	0.914	n.s.

Improvement over the default is oriented so that positive values are better (default – method for RMSE and MAE; method – default for R², ROC-AUC and PR-AUC). CIs: percentile bootstrap, 10,000 resamples of the 30 tuning seeds. p: one-sample Wilcoxon signed-rank test against the default value (zero_method = pratt), Holm correction over the 5 optimizers within each dataset × experiment × metric. Conclusion: better / worse than the default at Holm p < 0.05, otherwise n.s. PR-AUC chance level (test prevalence): BBBP 0.740, BACE 0.342.

Table S2. Friedman tests across the five optimizers (default excluded).

Dataset	Exp.	Metric	χ^2	p	Mean rank Random	Mean rank TPE	Mean rank GWO	Mean rank WOA	Mean rank PSO
ESOL	A	RMSE (primary)	7.63	0.106	3.50	3.13	2.70	2.50	3.17
ESOL	A	MAE	1.47	0.833	3.13	3.23	2.83	2.87	2.93
ESOL	A	R2	7.63	0.106	3.50	3.13	2.70	2.50	3.17
ESOL	B	RMSE (primary)	6.40	0.171	3.37	2.50	3.07	3.30	2.77
ESOL	B	MAE	8.99	0.0614	3.27	2.43	3.17	3.47	2.67
ESOL	B	R2	6.40	0.171	3.37	2.50	3.07	3.30	2.77
FreeSolv	A	RMSE (primary)	13.5	0.00900	3.37	3.07	2.27	3.60	2.70
FreeSolv	A	MAE	9.89	0.0423	3.27	3.27	2.40	3.43	2.63
FreeSolv	A	R2	13.5	0.00900	3.37	3.07	2.27	3.60	2.70
FreeSolv	B	RMSE (primary)	3.39	0.495	3.07	2.87	2.77	3.43	2.87
FreeSolv	B	MAE	4.05	0.399	3.03	2.97	2.83	3.47	2.70
FreeSolv	B	R2	3.39	0.495	3.07	2.87	2.77	3.43	2.87
Lipophilicity	A	RMSE (primary)	5.42	0.247	3.50	3.00	2.60	2.82	3.08
Lipophilicity	A	MAE	5.47	0.242	3.50	2.60	2.80	3.05	3.05
Lipophilicity	A	R2	5.42	0.247	3.50	3.00	2.60	2.82	3.08
Lipophilicity	B	RMSE (primary)	3.28	0.512	3.28	3.20	2.63	3.02	2.87
Lipophilicity	B	MAE	1.29	0.863	3.22	2.90	3.10	2.98	2.80
Lipophilicity	B	R2	3.28	0.512	3.28	3.20	2.63	3.02	2.87
BBBP	A	ROC-AUC (primary)	8.99	0.0614	3.15	3.33	2.25	3.08	3.18
BBBP	A	PR-AUC	8.08	0.0886	2.85	3.23	2.35	3.25	3.32
BBBP	B	ROC-AUC (primary)	11.4	0.0220	3.40	2.45	2.77	3.62	2.77
BBBP	B	PR-AUC	5.25	0.262	3.33	2.80	2.63	3.37	2.87
BACE	A	ROC-AUC (primary)	5.68	0.225	3.48	3.12	2.80	3.02	2.58
BACE	A	PR-AUC	4.35	0.360	3.02	3.48	2.98	2.72	2.80
BACE	B	ROC-AUC (primary)	6.26	0.181	3.23	2.93	3.17	3.27	2.40
BACE	B	PR-AUC	4.97	0.291	2.87	2.57	3.30	3.33	2.93

Friedman test with the 30 tuning seeds as blocks and the 5 optimizers as treatments (4 degrees of freedom); mean rank 1 = best.

Table S3. Pairwise comparisons of optimizers on the primary metric: Wilcoxon signed-rank test with Holm correction and Vargha–Delaney A12.

Dataset	Exp.	Metric	Pair (first vs second)	Median paired difference	p (raw)	p (Holm)	A12
ESOL	A	RMSE	Random vs TPE	-0.00591	0.349	1.00	0.451
ESOL	A	RMSE	Random vs GWO	-0.00715	0.0803	0.803	0.391
ESOL	A	RMSE	Random vs WOA	-0.0107	0.0919	0.828	0.346
ESOL	A	RMSE	Random vs PSO	-0.00455	0.213	1.00	0.399
ESOL	A	RMSE	TPE vs GWO	-0.00605	0.339	1.00	0.444
ESOL	A	RMSE	TPE vs WOA	-0.0128	0.0919	0.828	0.407
ESOL	A	RMSE	TPE vs PSO	-0.00428	0.428	1.00	0.469
ESOL	A	RMSE	GWO vs WOA	-0.00294	0.715	1.00	0.442
ESOL	A	RMSE	GWO vs PSO	0.00277	0.490	1.00	0.519
ESOL	A	RMSE	WOA vs PSO	0.00393	0.477	1.00	0.558
ESOL	B	RMSE	Random vs TPE	-0.0210	0.0327	0.294	0.340
ESOL	B	RMSE	Random vs GWO	-0.0158	0.416	1.00	0.473
ESOL	B	RMSE	Random vs WOA	-0.0116	0.700	1.00	0.474
ESOL	B	RMSE	Random vs PSO	-0.00465	0.253	1.00	0.397
ESOL	B	RMSE	TPE vs GWO	0.0142	0.0577	0.462	0.653
ESOL	B	RMSE	TPE vs WOA	0.0224	0.0113	0.113	0.636
ESOL	B	RMSE	TPE vs PSO	0.00277	0.371	1.00	0.549
ESOL	B	RMSE	GWO vs WOA	-0.00296	0.935	1.00	0.496
ESOL	B	RMSE	GWO vs PSO	-0.00668	0.318	1.00	0.419
ESOL	B	RMSE	WOA vs PSO	-0.0158	0.146	1.00	0.422
FreeSolv	A	RMSE	Random vs TPE	-0.0320	0.465	1.00	0.429
FreeSolv	A	RMSE	Random vs GWO	-0.146	0.00501	0.0451	0.328
FreeSolv	A	RMSE	Random vs WOA	0.0321	1.00	1.00	0.476
FreeSolv	A	RMSE	Random vs PSO	-0.113	0.124	0.745	0.358
FreeSolv	A	RMSE	TPE vs GWO	-0.0965	0.0549	0.384	0.372
FreeSolv	A	RMSE	TPE vs WOA	0.0531	0.360	1.00	0.606
FreeSolv	A	RMSE	TPE vs PSO	-0.0743	0.177	0.886	0.437
FreeSolv	A	RMSE	GWO vs WOA	0.133	0.00104	0.0104	0.701
FreeSolv	A	RMSE	GWO vs PSO	0.0800	0.360	1.00	0.579
FreeSolv	A	RMSE	WOA vs PSO	-0.0708	0.0310	0.248	0.322
FreeSolv	B	RMSE	Random vs TPE	-0.0356	0.655	1.00	0.530
FreeSolv	B	RMSE	Random vs GWO	-0.0370	0.685	1.00	0.463
FreeSolv	B	RMSE	Random vs WOA	0.0184	0.393	1.00	0.598
FreeSolv	B	RMSE	Random vs PSO	-0.0399	0.382	1.00	0.467
FreeSolv	B	RMSE	TPE vs GWO	-0.000924	0.641	1.00	0.420
FreeSolv	B	RMSE	TPE vs WOA	0.0529	0.529	1.00	0.553
FreeSolv	B	RMSE	TPE vs PSO	-0.0132	0.382	1.00	0.427
FreeSolv	B	RMSE	GWO vs WOA	0.129	0.0767	0.767	0.608
FreeSolv	B	RMSE	GWO vs PSO	0.0221	0.903	1.00	0.503
FreeSolv	B	RMSE	WOA vs PSO	-0.0353	0.119	1.00	0.353
Lipophilicity	A	RMSE	Random vs TPE	-0.00434	0.0577	0.473	0.351
Lipophilicity	A	RMSE	Random vs GWO	-0.00498	0.0473	0.473	0.358
Lipophilicity	A	RMSE	Random vs WOA	-0.00472	0.0473	0.473	0.417
Lipophilicity	A	RMSE	Random vs PSO	-0.00157	0.626	1.00	0.428
Lipophilicity	A	RMSE	TPE vs GWO	-0.00111	0.700	1.00	0.538
Lipophilicity	A	RMSE	TPE vs WOA	-0.00134	0.715	1.00	0.553
Lipophilicity	A	RMSE	TPE vs PSO	-0.000267	0.570	1.00	0.570
Lipophilicity	A	RMSE	GWO vs WOA	0.00227	0.808	1.00	0.524
Lipophilicity	A	RMSE	GWO vs PSO	0.000562	0.556	1.00	0.542
Lipophilicity	A	RMSE	WOA vs PSO	0.000557	0.821	1.00	0.515
Lipophilicity	B	RMSE	Random vs TPE	0.00128	0.685	1.00	0.528
Lipophilicity	B	RMSE	Random vs GWO	-0.00594	0.452	1.00	0.420
Lipophilicity	B	RMSE	Random vs WOA	-0.00168	0.984	1.00	0.497
Lipophilicity	B	RMSE	Random vs PSO	-0.00393	0.612	1.00	0.438
Lipophilicity	B	RMSE	TPE vs GWO	-0.00249	0.309	1.00	0.406
Lipophilicity	B	RMSE	TPE vs WOA	0.00426	0.984	1.00	0.486
Lipophilicity	B	RMSE	TPE vs PSO	-0.00227	0.452	1.00	0.426
Lipophilicity	B	RMSE	GWO vs WOA	0.00167	0.328	1.00	0.558
Lipophilicity	B	RMSE	GWO vs PSO	0.00294	0.761	1.00	0.519
Lipophilicity	B	RMSE	WOA vs PSO	0.000323	0.670	1.00	0.449
BBBP	A	ROC-AUC	Random vs TPE	0.00166	0.968	1.00	0.513
BBBP	A	ROC-AUC	Random vs GWO	-0.00925	0.00662	0.0662	0.281
BBBP	A	ROC-AUC	Random vs WOA	0.000187	0.853	1.00	0.488
BBBP	A	ROC-AUC	Random vs PSO	0.000937	0.967	1.00	0.520
BBBP	A	ROC-AUC	TPE vs GWO	-0.00784	0.00930	0.0837	0.297

Dataset	Exp.	Metric	Pair (first vs second)	Median paired difference	p (raw)	p (Holm)	A12
BBBP	A	ROC-AUC	TPE vs WOA	-0.000531	0.700	1.00	0.476
BBBP	A	ROC-AUC	TPE vs PSO	-0.00191	0.730	1.00	0.478
BBBP	A	ROC-AUC	GWO vs WOA	0.00868	0.0166	0.132	0.662
BBBP	A	ROC-AUC	GWO vs PSO	0.0116	0.0190	0.133	0.714
BBBP	A	ROC-AUC	WOA vs PSO	0.00141	0.789	1.00	0.518
BBBP	B	ROC-AUC	Random vs TPE	-0.00943	0.0699	0.489	0.363
BBBP	B	ROC-AUC	Random vs GWO	-0.00394	0.158	0.935	0.400
BBBP	B	ROC-AUC	Random vs WOA	0.00381	0.465	1.00	0.549
BBBP	B	ROC-AUC	Random vs PSO	-0.00644	0.156	0.935	0.364
BBBP	B	ROC-AUC	TPE vs GWO	0.00144	0.570	1.00	0.512
BBBP	B	ROC-AUC	TPE vs WOA	0.00450	0.00622	0.0622	0.667
BBBP	B	ROC-AUC	TPE vs PSO	0.000687	0.934	1.00	0.478
BBBP	B	ROC-AUC	GWO vs WOA	0.00725	0.0364	0.291	0.643
BBBP	B	ROC-AUC	GWO vs PSO	-0.00281	0.761	1.00	0.457
BBBP	B	ROC-AUC	WOA vs PSO	-0.00637	0.0310	0.279	0.327
BACE	A	ROC-AUC	Random vs TPE	-0.00356	0.813	1.00	0.468
BACE	A	ROC-AUC	Random vs GWO	-0.00490	0.0332	0.332	0.389
BACE	A	ROC-AUC	Random vs WOA	-0.00356	0.434	1.00	0.444
BACE	A	ROC-AUC	Random vs PSO	-0.00942	0.0878	0.790	0.366
BACE	A	ROC-AUC	TPE vs GWO	0.000385	0.537	1.00	0.445
BACE	A	ROC-AUC	TPE vs WOA	-0.000481	0.886	1.00	0.464
BACE	A	ROC-AUC	TPE vs PSO	-0.00240	0.129	1.00	0.387
BACE	A	ROC-AUC	GWO vs WOA	0.00183	0.428	1.00	0.537
BACE	A	ROC-AUC	GWO vs PSO	-0.00144	0.459	1.00	0.455
BACE	A	ROC-AUC	WOA vs PSO	-0.00269	0.388	1.00	0.416
BACE	B	ROC-AUC	Random vs TPE	-0.0000962	0.711	1.00	0.459
BACE	B	ROC-AUC	Random vs GWO	0.00163	0.644	1.00	0.437
BACE	B	ROC-AUC	Random vs WOA	-0.00125	0.777	1.00	0.471
BACE	B	ROC-AUC	Random vs PSO	-0.00875	0.0407	0.407	0.366
BACE	B	ROC-AUC	TPE vs GWO	0.000481	0.829	1.00	0.482
BACE	B	ROC-AUC	TPE vs WOA	0.00529	0.704	1.00	0.504
BACE	B	ROC-AUC	TPE vs PSO	-0.000673	0.304	1.00	0.399
BACE	B	ROC-AUC	GWO vs WOA	-0.000385	0.792	1.00	0.541
BACE	B	ROC-AUC	GWO vs PSO	-0.00673	0.221	1.00	0.406
BACE	B	ROC-AUC	WOA vs PSO	-0.00654	0.0407	0.407	0.376

Paired Wilcoxon signed-rank test over 30 seeds (zero_method = pratt), Holm correction over the 10 pairs within each dataset × experiment. Median difference and A12 are oriented so that values > 0 and > 0.5, respectively, favour the first optimizer of the pair. Secondary metrics (MAE, R², PR-AUC) are in 2_pairwise.csv; 4 of 160 secondary comparisons are Holm-significant.

Table S4. Validation–test gap and Spearman correlation between best validation and test scores.

Dataset	Exp.	Method	Metric	Median best validation	Median test	Median gap (test – val)	Spearman ρ [95% CI]
ESOL	A	Default	RMSE	0.861	0.889	0.0283	–
ESOL	A	Random	RMSE	0.731	0.924	0.190	-0.134 [-0.444, 0.205]
ESOL	A	TPE	RMSE	0.713	0.919	0.204	-0.0216 [-0.409, 0.370]
ESOL	A	GWO	RMSE	0.709	0.916	0.207	-0.114 [-0.507, 0.287]
ESOL	A	WOA	RMSE	0.729	0.911	0.181	0.0785 [-0.292, 0.425]
ESOL	A	PSO	RMSE	0.718	0.916	0.195	-0.165 [-0.503, 0.220]
ESOL	B	Default	RMSE	0.861	0.889	0.0283	–
ESOL	B	Random	RMSE	0.734	0.921	0.182	-0.172 [-0.536, 0.217]
ESOL	B	TPE	RMSE	0.713	0.900	0.182	0.187 [-0.229, 0.547]
ESOL	B	GWO	RMSE	0.713	0.919	0.204	-0.0363 [-0.397, 0.360]
ESOL	B	WOA	RMSE	0.737	0.918	0.188	0.0483 [-0.347, 0.389]
ESOL	B	PSO	RMSE	0.721	0.905	0.187	-0.157 [-0.510, 0.240]
FreeSolv	A	Default	RMSE	3.27	2.81	-0.462	–
FreeSolv	A	Random	RMSE	2.66	2.61	-0.0914	0.101 [-0.338, 0.484]
FreeSolv	A	TPE	RMSE	2.42	2.51	0.117	0.175 [-0.200, 0.492]
FreeSolv	A	GWO	RMSE	2.49	2.46	-0.0774	0.422 [0.0714, 0.722]
FreeSolv	A	WOA	RMSE	2.53	2.60	0.0609	0.226 [-0.204, 0.619]
FreeSolv	A	PSO	RMSE	2.62	2.49	-0.134	0.128 [-0.286, 0.517]
FreeSolv	B	Default	RMSE	3.27	2.81	-0.462	–
FreeSolv	B	Random	RMSE	2.41	2.46	-0.00343	0.0541 [-0.314, 0.423]
FreeSolv	B	TPE	RMSE	2.14	2.46	0.371	0.243 [-0.119, 0.547]
FreeSolv	B	GWO	RMSE	2.04	2.43	0.372	0.280 [-0.0982, 0.580]
FreeSolv	B	WOA	RMSE	2.41	2.54	0.144	-0.319 [-0.657, 0.0850]
FreeSolv	B	PSO	RMSE	2.28	2.39	0.165	-0.188 [-0.509, 0.199]
Lipophilicity	A	Default	RMSE	0.737	0.760	0.0231	–
Lipophilicity	A	Random	RMSE	0.699	0.729	0.0313	0.0354 [-0.318, 0.368]
Lipophilicity	A	TPE	RMSE	0.693	0.725	0.0322	-0.0897 [-0.438, 0.288]
Lipophilicity	A	GWO	RMSE	0.692	0.726	0.0371	-0.277 [-0.612, 0.142]
Lipophilicity	A	WOA	RMSE	0.697	0.727	0.0302	0.00111 [-0.385, 0.382]
Lipophilicity	A	PSO	RMSE	0.693	0.728	0.0352	0.260 [-0.151, 0.623]
Lipophilicity	B	Default	RMSE	0.737	0.760	0.0231	–
Lipophilicity	B	Random	RMSE	0.712	0.738	0.0285	0.220 [-0.151, 0.527]
Lipophilicity	B	TPE	RMSE	0.698	0.738	0.0430	0.0527 [-0.302, 0.376]
Lipophilicity	B	GWO	RMSE	0.697	0.734	0.0379	0.391 [0.0366, 0.673]
Lipophilicity	B	WOA	RMSE	0.705	0.739	0.0347	0.319 [-0.119, 0.700]
Lipophilicity	B	PSO	RMSE	0.698	0.734	0.0378	0.332 [-0.0834, 0.652]
BBBP	A	Default	ROC-AUC	0.924	0.915	-0.00927	–
BBBP	A	Random	ROC-AUC	0.944	0.903	-0.0405	-0.420 [-0.673, -0.0763]
BBBP	A	TPE	ROC-AUC	0.948	0.900	-0.0474	-0.283 [-0.625, 0.139]
BBBP	A	GWO	ROC-AUC	0.948	0.909	-0.0380	-0.00245 [-0.381, 0.377]
BBBP	A	WOA	ROC-AUC	0.946	0.904	-0.0460	-0.201 [-0.575, 0.206]
BBBP	A	PSO	ROC-AUC	0.945	0.901	-0.0442	-0.0307 [-0.394, 0.346]
BBBP	B	Default	ROC-AUC	0.924	0.915	-0.00927	–
BBBP	B	Random	ROC-AUC	0.948	0.904	-0.0434	0.0732 [-0.264, 0.394]
BBBP	B	TPE	ROC-AUC	0.952	0.911	-0.0423	0.168 [-0.248, 0.568]
BBBP	B	GWO	ROC-AUC	0.953	0.907	-0.0458	-0.0884 [-0.431, 0.268]
BBBP	B	WOA	ROC-AUC	0.950	0.904	-0.0456	-0.242 [-0.589, 0.179]
BBBP	B	PSO	ROC-AUC	0.950	0.909	-0.0401	0.0512 [-0.289, 0.379]
BACE	A	Default	ROC-AUC	0.859	0.814	-0.0450	–
BACE	A	Random	ROC-AUC	0.886	0.810	-0.0735	-0.449 [-0.727, -0.0984]
BACE	A	TPE	ROC-AUC	0.890	0.814	-0.0744	0.0915 [-0.290, 0.452]
BACE	A	GWO	ROC-AUC	0.887	0.815	-0.0704	-0.156 [-0.537, 0.280]
BACE	A	WOA	ROC-AUC	0.886	0.816	-0.0695	0.0238 [-0.306, 0.366]
BACE	A	PSO	ROC-AUC	0.886	0.819	-0.0683	-0.141 [-0.513, 0.270]
BACE	B	Default	ROC-AUC	0.859	0.814	-0.0450	–
BACE	B	Random	ROC-AUC	0.893	0.813	-0.0773	-0.0690 [-0.488, 0.378]
BACE	B	TPE	ROC-AUC	0.901	0.818	-0.0833	0.00979 [-0.382, 0.421]
BACE	B	GWO	ROC-AUC	0.904	0.819	-0.0896	-0.106 [-0.440, 0.240]
BACE	B	WOA	ROC-AUC	0.895	0.812	-0.0772	-0.0548 [-0.410, 0.319]
BACE	B	PSO	ROC-AUC	0.900	0.820	-0.0747	0.119 [-0.268, 0.484]

Validation score = best validation score of the run (for the default: its validation score), on the same scale as the test metric (Experiment B without the feature-count penalty). ρ : Spearman correlation between validation and test scores across the 30 seeds, percentile bootstrap CI (10,000 resamples); not defined for the deterministic default.

Table S5. Selected-feature fraction and Nogueira stability (Experiment B).

Dataset	F	Method	Selected fraction, median [95% CI]	IQR	Nogueira Φ [95% CI]
ESOL	188	Random	0.516 [0.487, 0.527]	0.473–0.540	0.0162 [0.00513, 0.0273]
ESOL	188	TPE	0.516 [0.503, 0.524]	0.491–0.532	0.0150 [0.00611, 0.0240]
ESOL	188	GWO	0.447 [0.431, 0.473]	0.422–0.473	0.0170 [0.00803, 0.0260]
ESOL	188	WOA	0.529 [0.476, 0.652]	0.473–0.705	-0.000654 [-0.0102, 0.00888]
ESOL	188	PSO	0.508 [0.495, 0.527]	0.489–0.540	0.0135 [0.00388, 0.0231]
FreeSolv	171	Random	0.515 [0.488, 0.538]	0.480–0.544	0.0235 [0.0121, 0.0349]
FreeSolv	171	TPE	0.485 [0.471, 0.497]	0.462–0.497	0.0264 [0.0170, 0.0359]
FreeSolv	171	GWO	0.430 [0.415, 0.442]	0.405–0.462	0.0257 [0.0145, 0.0369]
FreeSolv	171	WOA	0.579 [0.409, 0.678]	0.377–0.708	-0.000720 [-0.0122, 0.0107]
FreeSolv	171	PSO	0.488 [0.474, 0.512]	0.468–0.526	0.0172 [0.00755, 0.0268]
Lipophilicity	207	Random	0.519 [0.507, 0.531]	0.504–0.536	0.00959 [0.00180, 0.0174]
Lipophilicity	207	TPE	0.512 [0.498, 0.531]	0.494–0.535	0.0181 [0.00891, 0.0273]
Lipophilicity	207	GWO	0.469 [0.459, 0.476]	0.441–0.478	0.0141 [0.00785, 0.0204]
Lipophilicity	207	WOA	0.589 [0.548, 0.674]	0.529–0.691	-0.00122 [-0.0113, 0.00890]
Lipophilicity	207	PSO	0.517 [0.502, 0.541]	0.502–0.546	0.0171 [0.00829, 0.0259]
BBBP	210	Random	0.514 [0.490, 0.524]	0.479–0.531	0.00690 [-0.00177, 0.0156]
BBBP	210	TPE	0.490 [0.481, 0.521]	0.481–0.532	0.0111 [0.00109, 0.0211]
BBBP	210	GWO	0.440 [0.421, 0.450]	0.415–0.470	0.0186 [0.00810, 0.0292]
BBBP	210	WOA	0.500 [0.455, 0.557]	0.415–0.599	0.00269 [-0.00689, 0.0123]
BBBP	210	PSO	0.500 [0.488, 0.505]	0.481–0.508	0.0106 [0.00139, 0.0198]
BACE	193	Random	0.487 [0.477, 0.505]	0.472–0.513	0.0123 [0.00216, 0.0224]
BACE	193	TPE	0.503 [0.484, 0.526]	0.472–0.532	0.00971 [-0.00321, 0.0226]
BACE	193	GWO	0.443 [0.430, 0.461]	0.417–0.466	0.0334 [0.0194, 0.0473]
BACE	193	WOA	0.430 [0.409, 0.484]	0.399–0.497	0.00471 [-0.00441, 0.0138]
BACE	193	PSO	0.497 [0.482, 0.513]	0.482–0.521	0.0214 [0.0109, 0.0318]

F: number of descriptors available after training-split filtering. Selected fraction = selected / F per run; CI: percentile bootstrap. Nogueira stability over the 30 selected subsets with its asymptotic 95% CI (Nogueira et al., 2018); $\Phi = 0$ corresponds to random selection.

Table S6. Mirrored-encoding robustness check (pre-registered tests, Experiment B).

Dataset	Outcome	Optimizer	Median normal	Median mirrored	Median paired difference	Seeds mirrored > normal	p (raw)	p (Holm)
FreeSolv	Selected fraction	Random	0.515	0.503	0.00292	15/30	0.459	1.00
FreeSolv	Selected fraction	TPE	0.485	0.500	0.0175	18/30	0.133	0.532
FreeSolv	Selected fraction	GWO	0.430	0.553	0.135	28/30	<0.001	<0.001
FreeSolv	Selected fraction	WOA	0.579	0.500	-0.0643	13/30	0.453	1.00
FreeSolv	Selected fraction	PSO	0.488	0.494	-0.00292	12/30	0.636	1.00
FreeSolv	Test RMSE	Random	2.46	2.43	0.0393	17/30	0.598	1.00
FreeSolv	Test RMSE	TPE	2.46	2.46	-0.0182	14/30	0.871	1.00
FreeSolv	Test RMSE	GWO	2.43	2.37	-0.0781	13/30	0.428	1.00
FreeSolv	Test RMSE	WOA	2.54	2.46	-0.0414	11/30	0.516	1.00
FreeSolv	Test RMSE	PSO	2.39	2.46	0.0942	18/30	0.245	1.00
BBBP	Selected fraction	Random	0.514	0.505	0.00238	15/30	0.975	1.00
BBBP	Selected fraction	TPE	0.490	0.502	-0.00476	12/30	0.491	1.00
BBBP	Selected fraction	GWO	0.440	0.552	0.102	29/30	<0.001	<0.001
BBBP	Selected fraction	WOA	0.500	0.464	-0.0738	11/30	0.0804	0.322
BBBP	Selected fraction	PSO	0.500	0.495	-0.0143	10/30	0.621	1.00
BBBP	Test ROC-AUC	Random	0.904	0.904	0.000125	15/30	0.853	1.00
BBBP	Test ROC-AUC	TPE	0.911	0.909	0.00334	18/30	0.551	1.00
BBBP	Test ROC-AUC	GWO	0.907	0.908	-0.00256	13/30	0.839	1.00
BBBP	Test ROC-AUC	WOA	0.904	0.903	0.00575	18/30	0.360	1.00
BBBP	Test ROC-AUC	PSO	0.909	0.910	0.001000	16/30	1.00	1.00

Mirrored encoding: every proposed point x is decoded as $1 - x$ (including the shared initial design). Paired Wilcoxon signed-rank test over 30 seeds (zero_method = pratt), difference = mirrored – normal, Holm correction over the 5 optimizers within each dataset $\times$ outcome, as pre-registered in results/xgb_mh/mirror/prediction.md before the run. Decision rule: GWO supports the prediction if its mirrored fraction is higher (Holm $p < 0.05$) and its median mirrored fraction exceeds 0.5 on both datasets (met: 4 of 4 conditions); Random (control) shows no change (met).

Table S7. Sensitivity analysis: a0 versus reference-faithful coefficient schedule for GWO and WOA.

Method	Dataset	Exp.	Metric	Median reference	Median a0	Median paired difference	p (raw)	p (Holm)	Unique evaluations reference / a0
GWO	ESOL	A	RMSE	0.916	0.913	-0.00226	0.465	1.00	100 / 91.0
GWO	ESOL	B	RMSE	0.919	0.911	-0.00810	0.0417	0.417	100 / 91.0
GWO	FreeSolv	A	RMSE	2.46	2.51	0.0534	0.128	1.00	100 / 91.0
GWO	FreeSolv	B	RMSE	2.43	2.40	-0.0317	0.855	1.00	100 / 91.0
GWO	Lipophilicity	A	RMSE	0.726	0.726	0.000223	0.626	1.00	100 / 91.0
GWO	Lipophilicity	B	RMSE	0.734	0.738	0.00591	0.217	1.00	100 / 91.0
GWO	BBBP	A	ROC-AUC	0.909	0.907	0.00219	0.165	1.00	100 / 91.0
GWO	BBBP	B	ROC-AUC	0.907	0.907	0.00356	0.178	1.00	100 / 91.0
GWO	BACE	A	ROC-AUC	0.815	0.815	-0.00202	0.530	1.00	100 / 91.0
GWO	BACE	B	ROC-AUC	0.819	0.814	0.00288	0.422	1.00	100 / 91.0
WOA	ESOL	A	RMSE	0.911	0.905	-0.00318	0.309	1.00	97.5 / 92.0
WOA	ESOL	B	RMSE	0.918	0.911	-0.00339	0.935	1.00	97.5 / 92.5
WOA	FreeSolv	A	RMSE	2.60	2.59	-0.00604	0.805	1.00	96.5 / 91.5
WOA	FreeSolv	B	RMSE	2.54	2.48	-0.0380	0.289	1.00	97.0 / 92.0
WOA	Lipophilicity	A	RMSE	0.727	0.727	-0.000701	0.681	1.00	97.0 / 93.0
WOA	Lipophilicity	B	RMSE	0.739	0.746	0.00595	0.100	1.00	97.5 / 92.0
WOA	BBBP	A	ROC-AUC	0.904	0.902	-0.000125	0.886	1.00	96.5 / 92.0
WOA	BBBP	B	ROC-AUC	0.904	0.903	-0.000875	0.629	1.00	97.0 / 92.0
WOA	BACE	A	ROC-AUC	0.816	0.815	-0.00519	0.188	1.00	96.0 / 92.0
WOA	BACE	B	ROC-AUC	0.812	0.813	-0.000385	0.894	1.00	97.5 / 93.0

a0: coefficient a reaches exactly 0 on the last evaluated update (GWO collapses to one point, wasting the last 9 calls; WOA re-evaluates the best position); reference: $a = 2 - t \cdot (2/T)$, last evaluated update $a = 0.4$. Paired difference oriented so that > 0 favours the reference schedule; paired Wilcoxon test (zero_method = pratt), Holm correction over the 10 dataset $\times$ experiment cells per method. Tuning-versus-default conclusions that change when a0 replaces the reference variants: 0 of 50.

Table S8. Compute cost per dataset and optimizer.

Dataset	Method	Median s per fitted call	Median (min) unique evaluations per run	Worker-hours, sleep removed	Worker-hours, raw	Calls dropped (> 600 s)
ESOL	Random	2.01	100 (100)	5.88	5.88	0
ESOL	TPE	3.57	100 (100)	10.2	10.2	0
ESOL	GWO	4.02	100 (100)	8.95	8.95	0
ESOL	WOA	2.89	97.5 (93)	7.40	7.40	0
ESOL	PSO	3.27	100 (100)	7.16	7.16	0
FreeSolv	Random	1.23	100 (100)	3.11	3.11	0
FreeSolv	TPE	0.758	100 (100)	1.89	1.89	0
FreeSolv	GWO	0.704	100 (100)	1.97	1.97	0
FreeSolv	WOA	0.892	97.0 (63)	2.06	30.4	10
FreeSolv	PSO	0.898	100 (100)	2.40	2.40	0
Lipophilicity	Random	8.68	100 (100)	23.9	23.9	0
Lipophilicity	TPE	11.0	100 (100)	28.2	28.2	0
Lipophilicity	GWO	10.2	100 (100)	22.2	22.2	0
Lipophilicity	WOA	10.2	97.0 (92)	22.7	22.7	0
Lipophilicity	PSO	11.1	100 (100)	22.8	22.8	0
BBBP	Random	0.919	100 (100)	2.10	2.10	0
BBBP	TPE	1.47	100 (100)	3.23	3.23	0
BBBP	GWO	1.98	100 (100)	3.55	3.56	0
BBBP	WOA	1.36	97.0 (92)	2.69	2.69	0
BBBP	PSO	1.43	100 (100)	2.79	2.79	0
BACE	Random	1.38	100 (100)	3.19	3.19	0
BACE	TPE	1.08	100 (100)	2.87	2.87	0
BACE	GWO	1.28	100 (100)	3.17	3.17	0
BACE	WOA	1.16	97.0 (92)	2.65	2.65	0
BACE	PSO	1.35	100 (100)	3.13	3.13	0
All	All	–	–	200	229	10

Primary runs only (60 runs per dataset $\times$ method: 2 experiments $\times$ 30 seeds). Time per call measured under 10 concurrent workers (Intel i7-10750H, 6 cores / 12 threads), one XGBoost thread per model. Calls longer than 600 s contain machine-sleep periods and were dropped (10 calls in 10 runs).

Table S9. Validation headroom versus test gain (exploratory, not pre-specified).

Dataset	Exp.	Default validation	Median best validation (5 optimizers)	Validation headroom (%)	Default test	Median test (5 optimizers)	Test gain (%)
ESOL	A	0.861	0.722	16.1	0.889	0.916	-3.11
ESOL	B	0.861	0.725	15.8	0.889	0.909	-2.32
FreeSolv	A	3.27	2.57	21.5	2.81	2.52	10.4
FreeSolv	B	3.27	2.27	30.6	2.81	2.46	12.6
Lipophilicity	A	0.737	0.694	5.78	0.760	0.727	4.35
Lipophilicity	B	0.737	0.702	4.72	0.760	0.737	3.02
BBBP	A	0.924	0.946	2.34	0.915	0.903	-1.28
BBBP	B	0.924	0.951	2.83	0.915	0.906	-1.04
BACE	A	0.859	0.886	3.14	0.814	0.815	0.0118
BACE	B	0.859	0.899	4.62	0.814	0.817	0.354

Exploratory, not pre-specified. Headroom and test gain are oriented so that > 0 means the optimizers are better than the default (RMSE: default – optimizers; ROC-AUC: optimizers – default) and are expressed relative to the default's value. Spearman correlation over the 10 cells: $\rho = 0.406$, $p = 0.244$ (relative); $\rho = 0.394$, $p = 0.260$ (raw units); no multiplicity correction.

Table S10. Data-curation check: duplicate molecules within and across the scaffold splits.

Dataset	Molecules	Duplicate groups (extra copies) train	val	test	Cross-split train–val: exact / variant groups (pairs)	train–test	val–test	Label notes, within-split duplicates
ESOL	1128	9 (9)	2 (2)	0	0 / 0 (0)	0 / 0 (0)	0 / 0 (0)	max $ \Delta \log S $ within groups: train 1.03, val 0.10
FreeSolv	642	0	0	0	0 / 0 (0)	0 / 0 (0)	0 / 0 (0)	–
Lipophilicity	4200	0	0	0	0 / 0 (0)	0 / 0 (0)	0 / 0 (0)	–
BBBP	2039	62 (66)	0	1 (1)	1 / 0 (0); labels 1/1 agree	0 / 0 (0)	0 / 0 (0)	11 train groups with conflicting classes
BACE	1513	0	0	0	0 / 2 (2); labels 2/2 agree	0 / 7 (8); labels 8/8 agree	0 / 2 (2); labels 2/2 agree	–

Read-only check on the exact molecules and split used in the study (analysis/final/curation). Duplicate: same standard InChIKey (RDKit 2026.03.6). Variant: same InChIKey first block (connectivity) but different full key, i.e., different stereochemistry and/or protonation state; of the 12 BACE variant pairs, 2 differ only in stereochemistry, 2 only in protonation and 8 in both. Such pairs can cross the Bemis–Murcko scaffold split because scaffolds were computed from the SMILES as written. Total: 1 exact cross-split duplicate (BBBP, train–val) and 11 variant groups (BACE); all cross-split labels agree.